ECO 202 — Practice Exam 3 — Version B

Worked solutions

Fall 2026 | Classes 10–13

Equivalent exact expressions and valid alternative reasoning are acceptable. For practice study, attempt the paper first, then compare reasoning, close these solutions, and reconstruct any missed method independently.

Questions 1–4: multiple choice

Question	Answer	Reason and distractor check
1	C	The jump is $3/4 - 1/4 = 1/2$. A is mass strictly below 2; B is cumulative mass through 2; D adds incompatible cumulative quantities.
2	C	Linearity and independent-sum variance give the stated exact moments. A needs extra distributional assumptions; B confuses variance with SD; D confuses unbiasedness with zero realized error.
3	A	The LLN gives concentration; the CLT supplies the limiting shape after centering and scaling. B makes an incorrect path guarantee, C changes the wrong distribution, and D confuses a limit with an exact finite result.
4	D	$\mathrm{Cov}(X, Y) = (1/2)(2)(3) = 3$, so the difference variance is $4 + 9 - 6 = 7$. A omits covariance, B treats correlation as covariance, and C uses the sum's sign.

5. A nonlinear transformation can preserve dependence

(a) The CDF includes the mass at its argument:

$$F_X(t) = \begin{cases} 0, & t < -1, \\ 1/4, & -1 \le t < 0, \\ 1/2, & 0 \le t < 1, \\ 1, & t \ge 1. \end{cases}$$

The moments are $\mathbb{E}[X] = -1/4+1/2 = 1/4$, $\mathbb{E}[X^2] = 1/4+1/2 = 3/4$, and $\mathrm{Var}(X) = 3/4-(1/4)^2 = 11/16$.

(b) $H = 0$ has probability $1/4$ and $H = 1$ has probability $3/4$. It is $H = \mathbf{1}\{X \neq 0\}$. Thus $\mathbb{E}[H] = 3/4$ and $\mathrm{Var}(H) = (3/4)(1/4) = 3/16$.

(c) Here $XH = X^3 = X$ on the support. Thus

$$\mathbb{E}[XH] = \frac{1}{4}, \qquad \mathrm{Cov}(X, H) = \frac{1}{4} - \frac{1}{4}\frac{3}{4} = \frac{1}{16}.$$

It follows that $\mathrm{Var}(X + H) = 11/16 + 3/16 + 2/16 = 1$. A direct check is that $X + H$ equals 0 with probability $1/2$ and 2 with probability $1/2$.

(d) For example, $\mathbb{P}(X = 0, H = 0) = 1/4$, whereas $\mathbb{P}(X = 0)\mathbb{P}(H = 0) = (1/4)^2 = 1/16$. This violates independence. Also $\mathbb{E}[XH] = 1/4$ differs from $\mathbb{E}[X]\mathbb{E}[H] = 3/16$; factoring the product expectation would incorrectly assume a property absent here.

6. Estimating a population maximum

(a) Independence gives $\mathbb{P}(M_2 = 0) = (1/2)^2 = 1/4$. Both draws must be at most 2 for the maximum to be at most 2, so $\mathbb{P}(M_2 \le 2) = (3/4)^2 = 9/16$. Subtracting the probability of maximum 0 and taking the remaining complement gives

Possible M_2	0	2	4
Probability	1/4	5/16	7/16

The probabilities sum to one.

(b) $\mathbb{E}[M_2] = 2(5/16) + 4(7/16) = 19/8$. Its bias for 4 is $19/8 - 4 = -13/8$. The maximum never exceeds 4 and is strictly below 4 with positive probability, so its expectation is below the target.

(c) Directly from the errors,

$$\mathrm{MSE}(M_2; 4) = (-4)^2 \frac{1}{4} + (-2)^2 \frac{5}{16} + 0^2 \frac{7}{16} = \frac{21}{4}.$$

Bias averages signed errors; MSE averages squared errors, capturing variability as well as systematic offset. As a check, $\mathrm{Var}(M_2) = 167/64$ and squared bias is $169/64$, which sum to $21/4$; these separate calculations are not required.

(d) A draw misses 4 with probability $3/4$, and independent draws all miss it with probability $(3/4)^n$. Hence $\mathbb{P}(M_n = 4) = 1 - (3/4)^n$. The statistic's distribution depends on the number of draws; a two-draw distribution cannot describe all sample sizes.

7. A discrete count and its continuous approximation

(a) $K \sim \mathsf{Binomial}(48, 1/4)$ exactly. Each indicator has mean $1/4$ and variance $3/16$. Linearity and independence yield

$$\mathbb{E}[K] = 12, \quad \mathrm{Var}(K) = 48(3/16) = 9, \quad \mathrm{SE}(K) = 3.$$

Dividing by 48 yields $\mathbb{E}[\widehat{P}] = 1/4$ and $\mathrm{SE}(\widehat{P}) = 3/48 = 1/16$. Expected successes are 12 and expected failures are 36; neither is very small, supporting an ordinary Normal approximation here.

(b) The continuous upper boundary for $K \le 17$ is 17.5. Therefore

$$z = \frac{17.5 - 12}{3} = \frac{11}{6}, \qquad \mathbb{P}(K \le 17) \approx \Phi(11/6) = 0.9666.$$

The equivalent exact event for the sample proportion is $\widehat{P} \le 17/48$. The half-unit adjustment belongs to the count scale; on the proportion scale the corrected continuous boundary would be $17.5/48$.

(c) The LLN states that $\widehat{P}_n$ concentrates around $p = 1/4$, in the sense that fixed positive deviations have probability tending to zero. The CLT states that $(\widehat{P}_n - p)/\sqrt{p(1-p)/n}$ approaches a standard Normal distribution. The unscaled error shrinks; the centered and scaled error retains a nondegenerate limiting distribution.

(d) Twelve is the mean of the sampling distribution over repeated groups of 48 independent draws, not a required count in any one group. A realized count is one integer from 0 through 48, while the expected count is fixed by n and p and need not in general be an integer.

8. Combining estimators and reporting uncertainty

(a) Linearity gives $\mathbb{E}[T_1 - 3T_2] = \theta_1 - 3\theta_2$, so the combination is unbiased for its stated target. Independence is not required for this expectation result.

(b) The covariance contributes with the sign of the two coefficients:

$$\mathrm{Var}(T_1 - 3T_2) = 4 + 9(9) + 2(1)(-3)(1) = 79.$$

MSE is also 79 because the bias is zero. Adding only $4 + 9(9) = 85$ drops the nonzero covariance term; the two estimators share uncertainty under this model.

(c) $\mathbb{E}[T_1^2] = \theta_1^2 + 4$, so $\mathbb{E}[T_1^2 + T_2] = \theta_1^2 + \theta_2 + 4$. The bias is 4. Subtracting it gives the unbiased estimator $T_1^2 + T_2 - 4$. No independence assumption is needed for this expectation calculation.

(d) The estimated standard error is $s/\sqrt{n} = 10/5 = 2$ dollars. This estimates the sample mean's repeated-sampling SD; it does not reveal the realized error $80 - \mu$. The population mean μ is unknown, so neither the signed error nor its absolute magnitude is established. The sample SD, estimated SE, and realized error are three different quantities.